



Hands-On With the Neo4j Graph Platform Session 5



Hands On Review

Additional Query Exercises

:guide hands-on/cypher-movies.html

1. Filtering queries using the WHERE clause (13 query exercises)
:guide hands-on/cypher-exercise_2.html
2. Working with patterns in queries (6 query exercises)
:guide hands-on/cypher-exercise_3.html
3. Working with Cypher data (*COLLECT*, *COUNT*, maps)
:guide hands-on/cypher-exercise_4.html
4. Controlling query processing (*WITH*)
:guide hands-on/cypher-exercise_5.html
5. Controlling returns returned (*RETURN*, *DISTINCT*, *ORDER BY LIMIT*)
:guide hands-on/cypher-exercise_6.html

Cypher Manual

<https://neo4j.com/docs/cypher-manual/current/introduction/>

Cypher Cheat Sheet

<https://neo4j.com/docs/cypher-cheat-sheet/5/auradb-enterprise/>



Class Setup

- Aura -unpause your instance (**Highly recommended**)
 - Neo4j Desktop -Start your DB
 - Neo4j Sandbox -make sure instance is started
- 

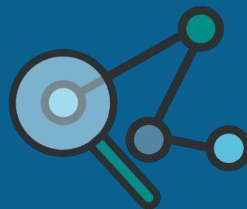
Class Agenda

- Labeled Property Graph (LPG) Components 🙌
- What is a graph database? 🙌
- What is Neo4j? 🙌
- Common Graph Use Cases 🙌

➤ **Graph modeling - Part 1** 🙌

- Graph modeling -Part 2 🙌
- Importing CSV into Neo4j
- Constraints and Indexes
- Intro to Graph Data Science
- Graph Data Science workflows

🙌 Hands-On

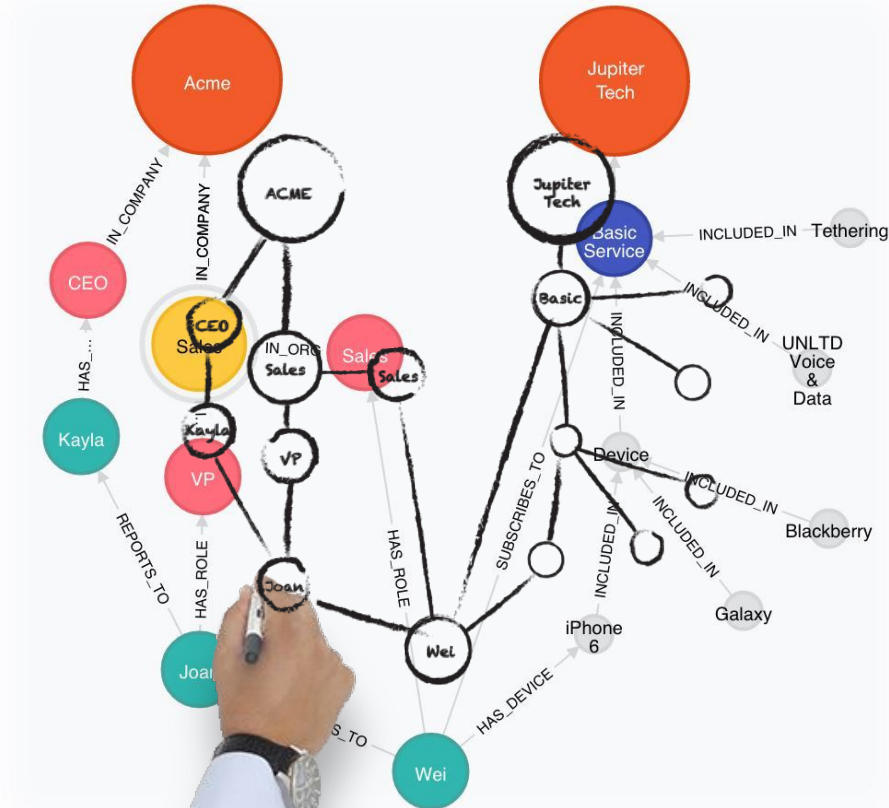


Modeling Your Data As A Graph

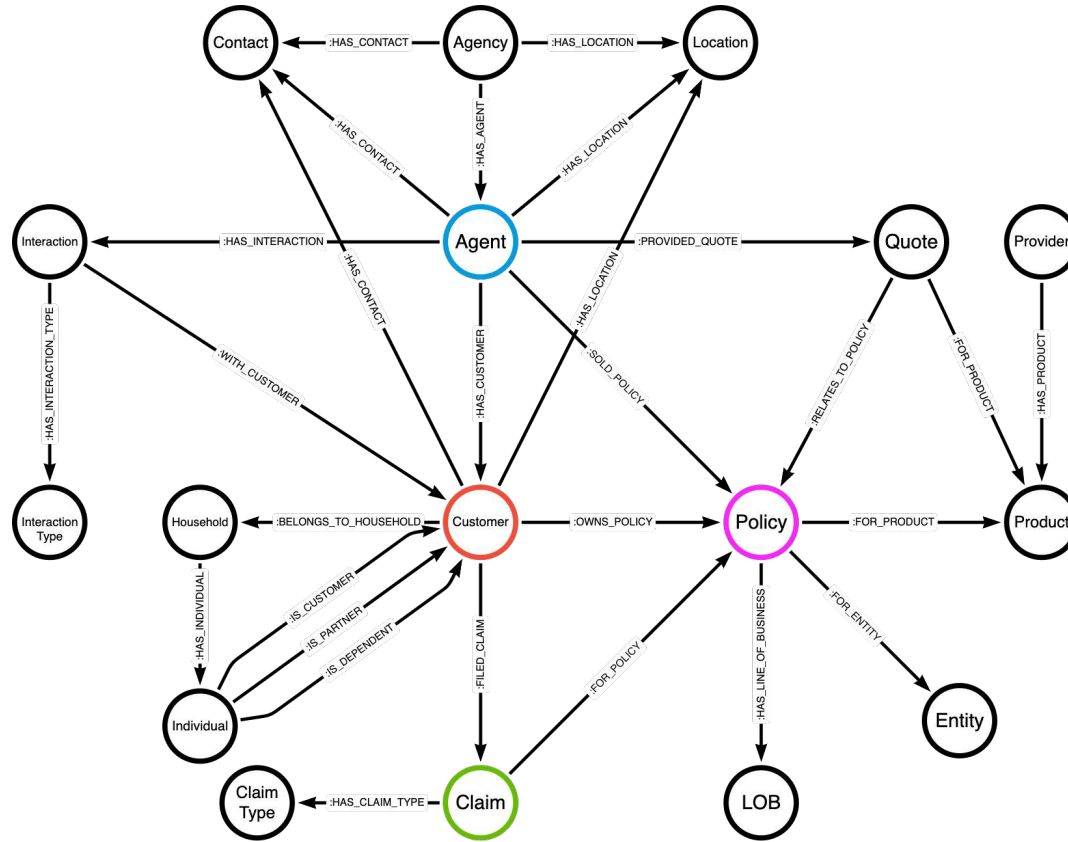
Neo4j Graph

- Look around your work area
- Think about “things” or “nouns” that could be in the graph.
- What are the attributes of that thing? Maybe age, height
- Think about “verbs” that could be the relationships between the “nouns”
- The verbs or relationships might have attribute, maybe distance, weight or confidence.
- And most importantly – you will notice that there is only one of you.

The Whiteboard Model Is the Graph Model

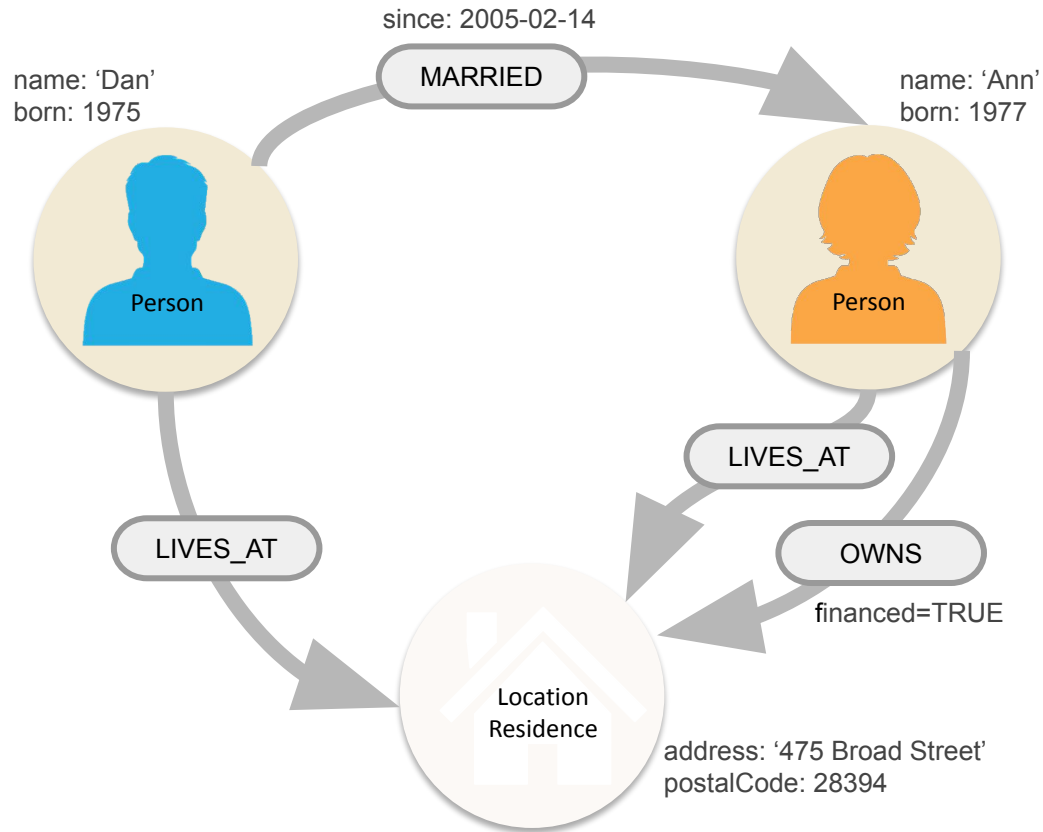


Insurance graph model from whiteboarding workshop

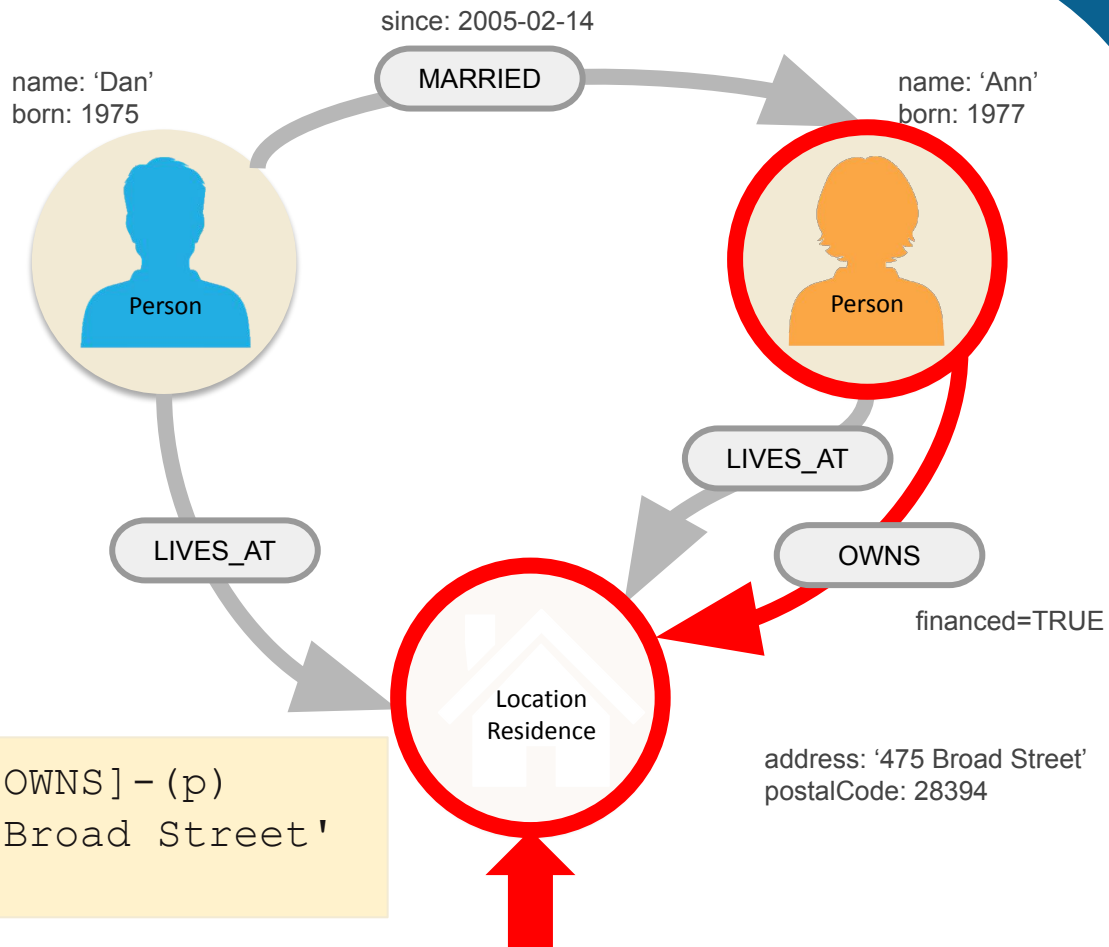


Neo4j Property Graph Model

- Nodes (Entities)
- Relationships
- Properties
- Labels

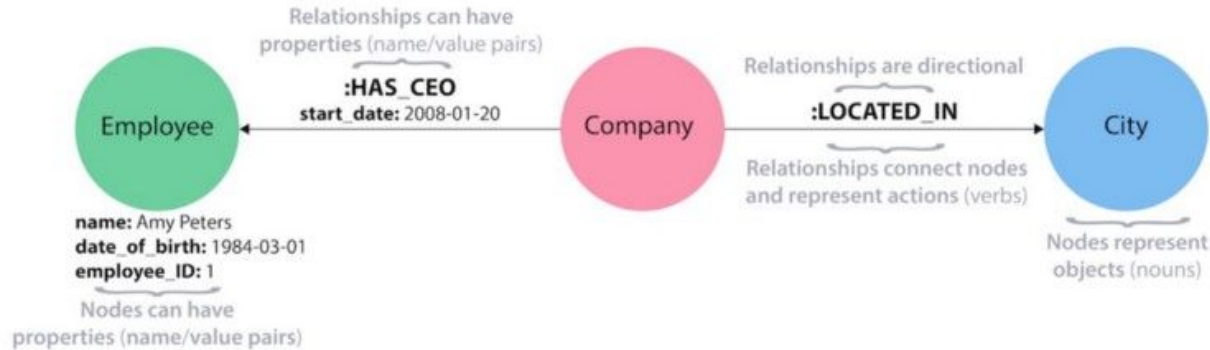


Traversal



```
MATCH (r:Residence)<-[:OWNS]-(p)
WHERE r.address = '475 Broad Street'
RETURN p
```

Graph Data Modeling: Categorical Variables

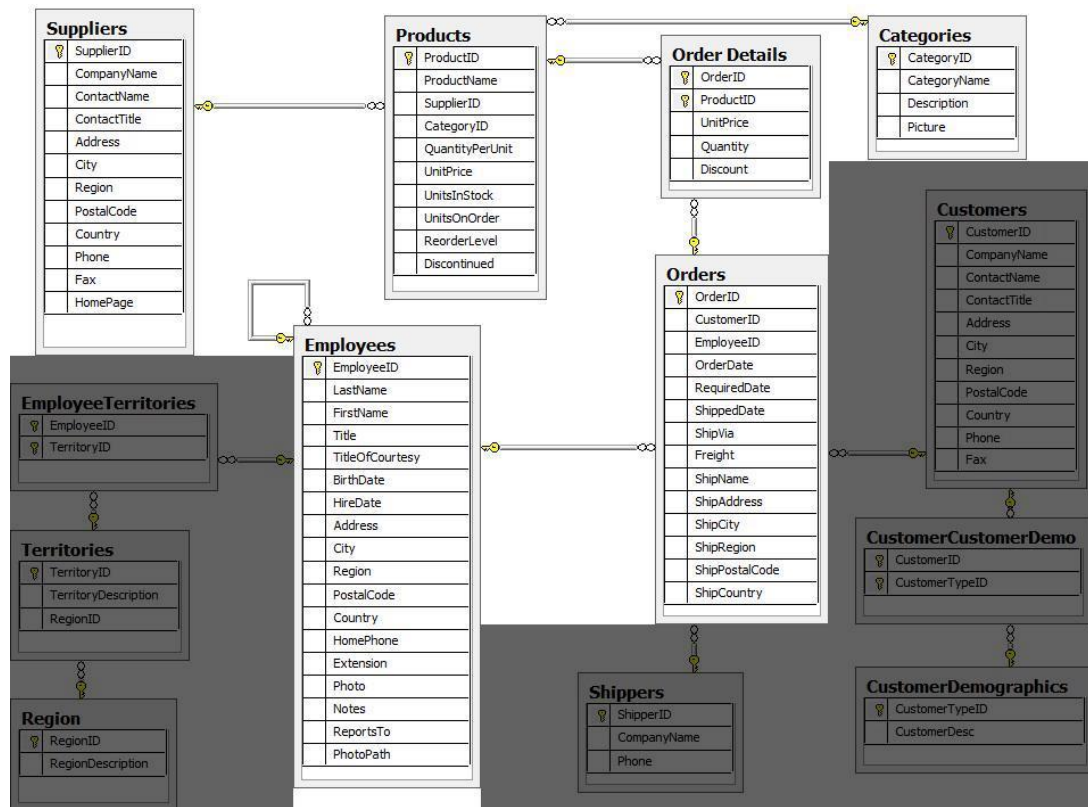


Data modeling is part art and part science; there are no hard and fast right answers, there's only what works well for your use case and your questions.

Section content from: <https://medium.com/neo4j/graph-data-modeling-categorical-variables-dd8a2845d5e0>

Modeling Your Data As A Graph Exercise

Exercise: Northwind Dataset



Data Modeling

When deriving a graph model from a relational model, you should keep a couple of general guidelines in mind.

- A row is a **node**
- A table name is a **label** name
- A join or foreign key is a **relationship**

Relational	Graph
Try and get the schema defined and then make minimal changes to it after that.	It's common for the schema to <i>evolve</i> with the application.
More abstract focus when modeling. i.e. Focus on classes rather than objects.	Common to use <i>actual data items</i> when modeling.

Modeling relational to graph

In some ways they're similar:

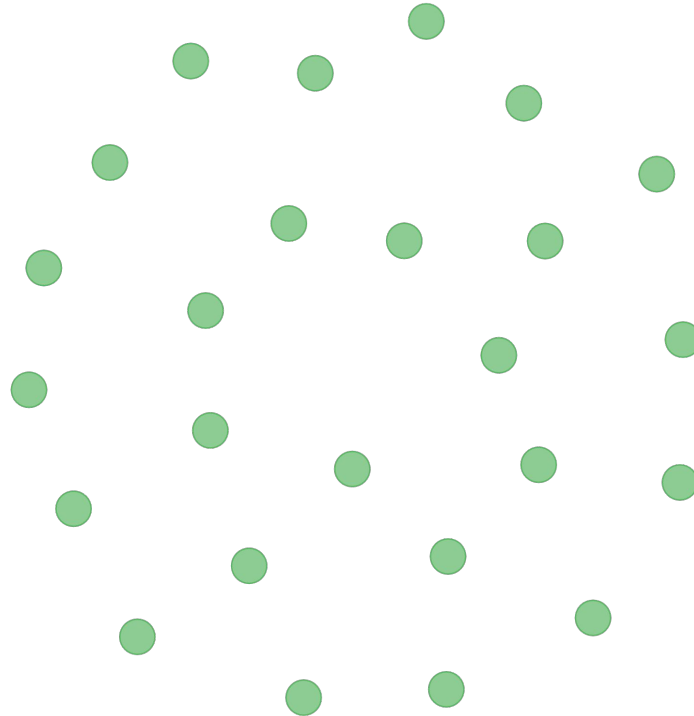
Relational	Graph
Rows	Nodes
Joins	Relationships
Table names	Labels
Columns	Properties

In some ways they're not:

Relational	Graph
Each column must have a field value.	Nodes with the same label aren't required to have the same set of properties.
Joins are calculated at query time.	Relationships are stored on disk when they are created.
A row can belong to one table.	A node can have many labels.

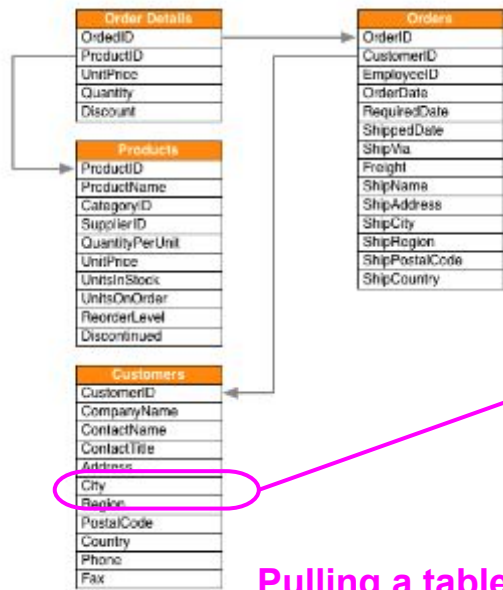
Rows to Nodes, Table Names to Labels

Orders	
🔑	OrderID
	CustomerID
	EmployeeID
	OrderDate
	RequiredDate
	ShippedDate
	ShipVia
	Freight
	ShipName
	ShipAddress
	ShipCity
	ShipRegion
	ShipPostalCode
	ShipCountry

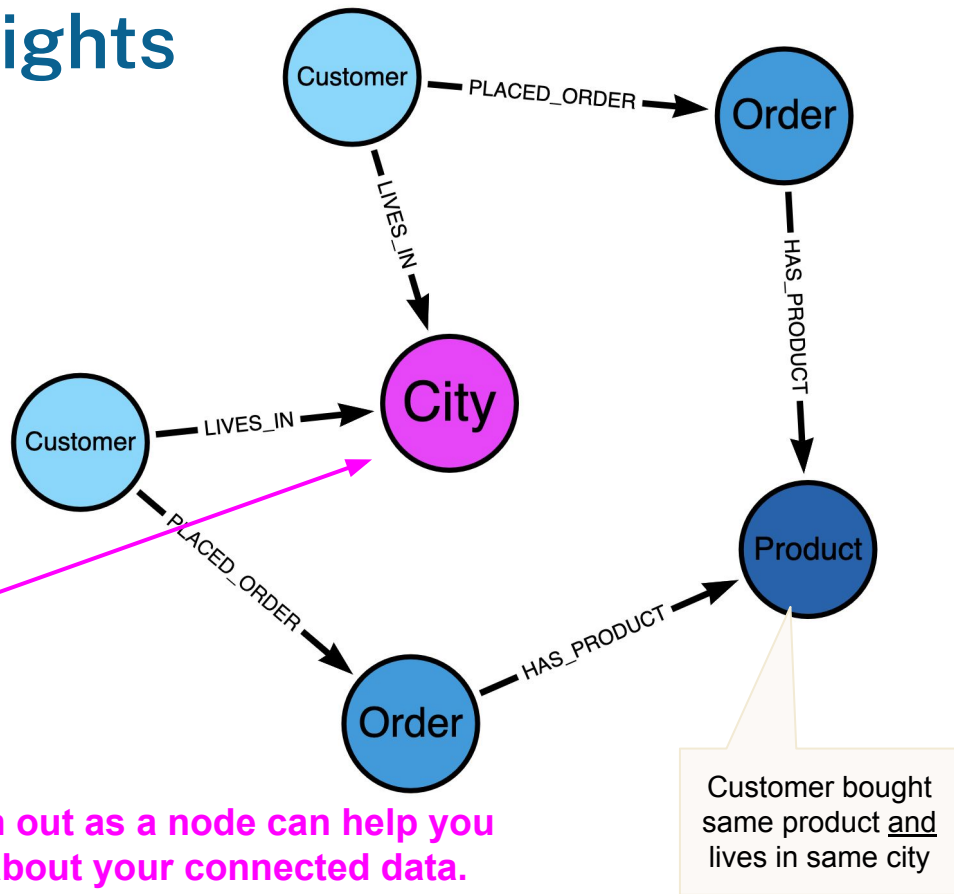


Modeling for better insights

Customer Orders



Pulling a table column out as a node can help you gain further insights about your connected data.



Write Your Queries First!

- Knowing the kinds of questions and queries you want to ask of your data is a great way to determine the structure of your data model.
 - If you know your queries need to return results within a certain date range, then you probably should ensure that date is not a property on a node, but rather stored as a separate node or relationship.
- Even if you do not know the exact query syntax just yet, **understanding the intention of the system or application you are building and then constructing the model around the business need** will help you organize it in a more accurate way.

Identify entities from the questions

Entities are the **nouns** in your application questions

- What ingredients are used in a recipe?

Ingredient

Recipe

- Who is married to this person?

Person

- The generic nouns will often become labels in your model
- Use domain knowledge to decide if entities need to be further grouped or differentiated

Identify entities from the questions

Properties must serve one of two purposes

- Unique identification
- Answering one of the application questions
 - Anchoring
 - Navigation
 - Output

Otherwise, they are merely decoration

Identify connections between entities

Connections are the **verbs** in your application questions

- What *ingredients* are used in a *recipe*?



- Who is married to this *person*?



Joins (in RDBMS) Become Relationships

- Join between **Supplier** and **Product** becomes a relationship named **SUPPLIES** (where supplier supplies product).
- Join between **Product** and **Category** becomes a relationship named **PART_OF** (where product is part of a category).
- Join between **Employee** and **Order** becomes a relationship named **SOLD** (where employee sold an order).
- Join between **Employee** and itself (unary relationship) becomes a relationship named **REPORTS_TO** (where employees have a manager).
- Join with join table (Order Details) between **Order** and **Product** becomes a relationship named **CONTAINS** with properties of unitPrice, quantity, and discount (where order contains a product).





Hands On Time!

Run the Guidebook

Run:

:guide hands-on

Select:

Modeling and Importing Northwind data **(Steps 1 to 10)**

:guide hands-on/modeling-northwind.html

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